<table>
<tr><th rowspan="2">No</th><th rowspan="2">Title</th><th rowspan="2">Author (Year)</th><th colspan="3">Variable</th><th rowspan="2">Source Link</th><th rowspan="2">Paper Claim</th></tr>
<tr><th>Control</th><th>Independent</th><th>Dependent</th></tr>
<tr><td colspan="8">Pillar 1: Non-Graph & Conventional Sequential Approaches</td></tr>
<tr><td>[1]</td><td>Anomaly Detection in Network Traffic Using Machine Learning</td><td>Verma et al. (2026)</td><td>Standard network traffic feature set; identical train/test dataset partitioning protocol used across all ML classifiers.</td><td>Type of supervised machine learning classifier applied (e.g., Random Forest, SVM, Decision Tree).</td><td>Anomaly detection accuracy, precision, recall, and F1-score on labeled network traffic.</td><td>https://doi.org/10.56726/IRJMETS90853</td><td>Low computational complexity; good baseline for sudden volumetric shifts.</td></tr>
<tr><td>[2]</td><td>Network Intrusion Detection Using Supervised Machine Learning For DDoS Attack Classification</td><td>Shireen & Tasneem (2026)</td><td>Fixed DDoS traffic dataset with consistent preprocessing and feature normalization across classifiers.</td><td>Supervised ML algorithm selected for DDoS variant classification.</td><td>Classification accuracy and false negative rate on pre-defined DDoS attack categories.</td><td>https://rjwave.org/ijedr/papers/IJEDR2601354.pdf</td><td>Achieves high accuracy for static, pre-defined DDoS classification variants.</td></tr>
<tr><td>[3]</td><td>Log Anomaly Detection Method Based on Transformer and Temporal Convolutional Networks</td><td>Liao & Liu (2025)</td><td>Same raw system log corpora and preprocessing tokenization pipeline applied uniformly across model variants.</td><td>Architecture configuration: standalone Transformer, standalone TCN, or combined Transformer+TCN pipeline.</td><td>Log anomaly detection F1-score and computational efficiency (latency, parameter count).</td><td>https://doi.org/10.1109/ACCESS.2025.3561669</td><td>Captures long-range semantic patterns inside unstructured system text logs.</td></tr>
<tr><td>[4]</td><td>Network Intrusion Detection and Cyber Attack Classification Using CNN-LSTM Deep Learning Model</td><td>V. B, Y. S, & Jacob (2026)</td><td>Identical network packet dataset and sliding-window segmentation applied to all tested architectures.</td><td>Model architecture: standalone CNN, standalone LSTM, or hybrid CNN-LSTM.</td><td>Intrusion detection accuracy, precision, recall, and latency on network packet classification.</td><td>https://doi.org/10.1109/ICSADL67539.2026.11451829</td><td>Extracts localized spatial feature arrays along with raw time-series sequences.</td></tr>
<tr><td colspan="8">Pillar 2: Static and Snapshot-Based Graph Neural Networks</td></tr>
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[5]	Network Anomaly Detection with Graph Neural Networks (UPC Doctoral Thesis)	Martínez et al. (2025)	Fixed network topology graph with consistent node feature encoding across GNN variants.	GNN architecture variant (GCN, GAT, GraphSAGE) applied to static network topology snapshots.	Node-level anomaly classification accuracy, AUC-ROC, and precision-recall on structured network graphs.	https://upcommons.upc.edu	Maps structural properties of fixed network layouts for robust node classification.
[6]	Real-Time Bluetooth Man-in-the-Middle Attack Detection System Based on Graph Neural Network	Zhang (2025)	Controlled Bluetooth communication environment with fixed master-slave pair configurations.	GNN model applied to Bluetooth interaction graph vs. non-graph baseline detector.	MitM attack detection rate, false positive rate, and real-time detection latency.	https://doi.org/10.36227/techrxiv.174002494.45429007/v1	Efficient at capturing direct master-slave connection changes in dynamic pairs.
[7]	Network-Based Anomaly Detection in Blockchain Transactions Using GNN and DBSCAN	Guballo (2026)	Fixed blockchain transaction dataset with consistent graph construction and DBSCAN hyperparameters.	Combination strategy: GNN alone vs. GNN + DBSCAN clustering.	Malicious transaction detection precision, recall, and cluster quality metrics.	https://doi.org/10.47738/jcrb.v3i1.55	Groups malicious transactions effectively by merging topological metrics with density sweeps.
[8]	Research on System Log Anomaly Detection Method Based on Graph Neural Network	Y. Wang (2025)	Identical system log corpus parsed using the same log parsing template across model variants.	Presence or absence of graph-based relational encoding in the log processing pipeline.	Log anomaly detection accuracy and inter-process dependency capture rate.	https://doi.org/10.1109/CVIDL65390.2025.11085475	Reconstructs relational connection maps directly from raw log strings.
[9]	GNN Graph Structures in Network Anomaly Detection	Cressant et al. (2025)	Same GNN model architecture and training protocol applied across all graph construction variants.	Graph construction design choices: edge definition, feature aggregation depth, temporal granularity.	Intrusion detection performance metrics (F1, precision, recall) across different graph topology designs.	https://doi.org/10.1109/NOMS57970.2025.11073640	Detailed study on how graph construction settings alter intrusion detection output.
[10]	Anomaly Detection in Network Security Using Unsupervised Graph Neural Network	Sathishkumar et al. (2025)	Fixed unlabeled network traffic graph; same unsupervised training objective applied across	Unsupervised GNN configuration variant and anomaly scoring strategy.	Anomaly detection rate and false alarm rate without reliance on ground-truth attack	https://doi.org/10.1109/ICoACT63339.2025.1100	Runs independently of manual attack labels by treating rare structures as threats.

			configurations.		labels.	4728	
[11]	Log-Based Anomaly Detection of System Logs Using Graph Neural Network	Alsalmi et al. (2026)	Same structured system call log dataset and entity-relation graph schema applied across model variants.	GNN architecture applied to typed entity-relation graphs derived from parsed log attributes.	Anomaly detection accuracy and processing speed on structured system call trace datasets.	https://doi.org/10.32604/cmc.2025.071012	Captures fine-grained metadata parameters from structured system call traces.
[12]	Authentication Graphs: Analyzing User Behavior Within an Enterprise Network	Kent et al. (2015)	Fixed enterprise authentication log dataset from the LANL environment.	Graph representation of entity authentication interactions vs. flat tabular event records.	Behavioral baselining quality and anomaly identification rate for enterprise user-host authentication pairs.	https://doi.org/10.1016/j.cose.2014.09.001	Builds highly solid historical behavior graphs for corporate identity tracking.
Pillar 3: Spatio-Temporal and Discrete Dynamic Graphs							
[13]	An Anomaly Aware Spatio-Temporal Graph Attention Network for Integrated Forecasting and Event Detection in Urban Traffic Streams	Bai & Thirumaran (2026)	Fixed urban traffic grid dataset with uniform time-window discretization parameters.	Inclusion of the spatio-temporal attention module vs. spatial-only or temporal-only GNN baseline.	Traffic anomaly event detection accuracy and spatial-temporal forecasting RMSE.	https://doi.org/10.2991/978-94-6239-616-6_112	Synchronously analyzes regional layout changes alongside temporal flow predictions.
[14]	Hierarchical Spatio-Temporal Graph Auto-encoded Federated Contrastive Learning For Attack Detection in VPN-Assisted Wireless IoT Network	Sangeetha & Sudha (2026)	Fixed VPN-assisted wireless IoT network topology and federated client configuration.	Contrastive learning objective applied within hierarchical spatio-temporal graph autoencoder vs. standard reconstruction loss.	Attack detection accuracy and communication overhead in federated wireless IoT environments.	https://doi.org/10.25258/ijddt.16.3s.18	Uses contrastive loss to secure wireless IoT devices across dynamic VPN tunnels.

[15]	Anomaly Detection Based Self-Healing Mechanism Using Dynamic Diffusion Spatio-Temporal Graph Convolutional Network in Industrial IoT	S. K et al. (2026)	Fixed industrial IoT sensor stream dataset and graph topology; consistent diffusion operator hyperparameters.	Addition of dynamic diffusion GCN layer and self-healing response module vs. standard ST-GCN.	Anomaly detection rate, false alarm rate, and self-healing response latency in industrial IoT environments.	https://doi.org/10.1016/j.knosys.2025.114812	Blends dynamic anomaly indexing with self-healing operational responses.
[16]	AddGraph: Anomaly Detection in Dynamic Graph Using Attention-based Temporal GCN	Zheng et al. (2019)	Fixed dynamic graph benchmark dataset; identical GCN depth and training epochs across ablation variants.	Inclusion of the attention mechanism within the temporal GCN vs. unweighted temporal aggregation.	Dynamic graph anomaly detection AUC-ROC and precision-recall scores.	https://arxiv.org/abs/1907.11976	Employs attention blocks to highlight sudden structural mutations over time.
[17]	Anomaly Detection in Dynamic Graphs via Transformer	Liu et al. (2023)	Fixed dynamic graph dataset; identical positional encoding and tokenization scheme across model variants.	Transformer-based multi-hop graph attention depth vs. shallower graph-based attention baselines.	Edge-level anomaly detection AUC-ROC and F1-score on dynamic graph benchmarks.	https://doi.org/10.1109/TKDE.2021.3124061	Models multi-hop connectivity updates smoothly using deep self-attention.
Pillar 4: Continuous-Time Dynamic Graphs & TGN Frameworks							
[18]	Temporal Graph Networks for Deep Learning on Dynamic Graphs	Rossi et al. (2020)	Standard dynamic graph benchmark datasets (Wikipedia, Reddit); fixed message aggregation protocol.	TGN memory module variant: no memory, GRU memory, or embedding memory with different message functions.	Temporal link prediction accuracy and inductive node classification performance on continuous-time dynamic graphs.	http://arxiv.org/abs/2006.10637	Pioneers real-time memory nodes and graph operators for instantaneous edge updates.
[19]	Federated Temporal Graph Learning for Weakly Supervised Bearing Anomaly Detection	Darlami & Awasthi (2026)	Fixed federated client partitioning and communication rounds; same weak supervision signal source.	Federated temporal graph learning with weak supervision vs. centralized fully-supervised TGN.	Bearing anomaly detection accuracy and label efficiency under federated weak supervision.	https://doi.org/10.64539/sjer.v2i2.2026.413	Reduces target label dependency using decentralized weak supervision paths.

[20]	Temporal Graph Neural Networks for Sequential Anomaly Detection in Real-Time E-Commerce Streams	Tan et al. (2025)	Fixed e-commerce transaction stream dataset; same TGN memory size and aggregation protocol.	TGN applied to buyer-item transaction graph vs. non-graph sequential fraud detection baseline.	Real-time fraud detection recall and false positive rate on high-speed transaction event streams.	https://doi.org/10.61784/asat3015	Tracks immediate fraud steps across high-speed transaction streams.
[21]	TGCAE: Temporal Graph Convolutional Autoencoder for Robust Anomaly Detection in Object-Centric Process Mining	Cao et al. (2026)	Fixed object-centric process log dataset; identical autoencoder reconstruction threshold calibration.	Temporal graph convolutional autoencoder architecture vs. standard process mining baselines.	Anomaly deviation score and reconstruction error on object-centric dynamic process logs.	https://doi.org/10.21203/rs.3.rs-8816450/v1	Reconstructs dynamic graph logs blindly to flag unmapped deviations.
[22]	Anomaly Detection in Containerized Tactical Systems Using Temporal Graph Neural Networks	Kwon et al. (2025)	Fixed containerized tactical system environment; identical TGN memory configuration across experiments.	TGN applied to dynamic microservice container interaction graphs vs. non-temporal graph baseline.	Isolation breach detection rate and response latency within ephemeral tactical container environments.	https://doi.org/10.1109/MILCOM64451.2025.11310523	Captures high-speed isolation breaches inside modern ephemeral tactical pods.
[23]	Granomaly: A Framework for Anomaly Detection in 5G Core Network Control Plane Traffic with Temporal Graph Neural Networks	Fritz et al. (2025)	Fixed 5G core network control plane traffic dataset; identical TGN memory and message function.	Granomaly TGN-based framework vs. conventional signature-based and non-temporal detection baselines.	Control plane anomaly detection precision, recall, and micro-temporal variance capture accuracy.	https://doi.org/10.1109/NOMS57970.2025.11073582	Captures precise micro-temporal variance inside 5G control plane traffic.
[24]	Protocol-Aware Anomaly Detection in IoT Networks Using Temporal Graph Neural Networks	Kundu et al. (2025)	Fixed IoT communication dataset; identical protocol encoding schema applied across model configurations.	Protocol-aware behavior rule injection into TGN layers vs. protocol-agnostic TGN baseline.	IoT anomaly detection F1-score and protocol violation identification rate.	https://doi.org/10.1109/CIACON65473.2025.11189365	Embeds specific network protocol behavior rules straight into the dynamic layers.

[25]	Real-Time Dynamic Multimodal Federated Anomaly Detection With Explainable Temporal Graph Learning	Mishra et al. (2026)	Fixed federated infrastructure topology; consistent multimodal feature fusion protocol across experiments.	Addition of explainability (XAI) module to the federated temporal graph learning pipeline.	Anomaly detection accuracy, explainability fidelity score, and privacy preservation in distributed environments.	https://doi.org/10.1109/ICCCES66261.2026.11436578	Blends strong privacy features with readable explainability paths for dynamic flags.
[26]	One-Class Intrusion Detection with Dynamic Graphs	Liuliakov et al. (2025)	Fixed normal-only training data from the same network environment; identical graph construction protocol.	One-class dynamic graph model vs. binary-class supervised TGN baseline.	Intrusion detection recall, false positive rate, and robustness under baseline behavioral variance.	http://arxiv.org/abs/2508.12885	Maps legitimate behavior bounds perfectly without requiring malicious data profiles.
[27]	Euler: Detecting Network Lateral Movement via Scalable Temporal Link Prediction	King & Huang (2023)	Fixed enterprise network authentication log dataset; consistent temporal link prediction objective.	Euler scalable CTDG framework vs. static graph and non-temporal lateral movement detectors.	Euler scalable CTDG framework vs. static graph and non-temporal lateral movement detectors.	https://github.com/iHeartGraph/Euler	Built to stop internal lateral movement by estimating dynamic link shifts.
[28]	CAPTOR: Cyber Attack Protection via Temporal Online Graph Representation Learning	Lakha et al. (2026)	Fixed streaming network traffic environment; identical graph representation update frequency.	CAPTOR online temporal graph learning vs. offline batch-trained temporal graph baselines.	Real-time attack detection recall, adaptation speed to new exploit patterns, and false positive rate.	https://doi.org/10.1109/TBDATA.2025.3621145	Instant streaming updates that adapt seamlessly to brand-new exploit patterns.
[29]	Graph Neural Networks for Anomaly Detection: A Systematic Review of Dynamic Temporal Approaches	Ares-Robledo et al. (2026)	Systematic literature review methodology (PRISMA); fixed inclusion/exclusion criteria applied uniformly.	GNN paradigm category: static GNNs vs. discrete-time dynamic GNNs vs. continuous-time dynamic GNNs.	Detection performance trend, architectural coverage, and identification of open research challenges across categories.	https://doi.org/10.1007/s10462-026-11532-7	Synthesizes comprehensive global dataset structures and open DyGAD limits.
[30]	BAG: Benchmarking Anomaly Detection on Dynamic Graphs	Hua et al. (2026)	Fixed benchmark protocol: 25 models evaluated on 10	Model paradigm: static GNN vs. discrete-time dynamic GNN vs.	Edge-level anomaly detection AUC-ROC and F1-score across all	https://arxiv.org/abs/2508.12885	Validates 25 models across 10 datasets, proving CTDG superiority on edge

			dynamic graph datasets with identical evaluation metrics.	continuous-time dynamic GNN.	10 benchmark datasets.		errors.